Table of contents

Part 6 — VPN privacy: Aura, Mullvad, and your options

> **The payoff of this part:** understand what a VPN actually does for you, see how the VPN you already use (Aura) compares to a privacy-focused option (Mullvad), and learn the Tailscale-native way to route traffic — the *exit node* — so you can pick the setup that fits. The messier question of how all this behaves *away from home* (and how it collides with your Pi-hole) gets its own treatment in Part 7.

Everything so far has been about reaching *into* your homelab. This part is the opposite direction: making your devices' traffic leave through a VPN, so the websites and apps you use can't see your real IP address, and so your ISP or a café's Wi-Fi can't see what you're doing.

You already use **Aura VPN** to change locations. That's a perfectly good consumer VPN. The goal here isn't to talk you out of it — it's to lay out the landscape clearly: what a VPN does, where Aura fits, where **Mullvad** fits, and the one Tailscale concept (the **exit node**) that ties VPNs into the homelab you've been building.

What a VPN does (and what it doesn't)

A VPN routes all your internet traffic through a remote server before it reaches the wider internet. Two practical consequences:

- **Your IP is hidden.** Sites see the VPN server's address, not yours — useful for privacy and for appearing to be in another location.
- **Your local network can't snoop.** On untrusted Wi-Fi, the airport or hotel can see only that you're talking to a VPN, not what you're doing.

What a VPN does *not* do: it isn't anonymity (the VPN provider can see your traffic, so provider trust matters), and it doesn't block ads by itself unless the provider offers DNS-level filtering. Keep that ad-blocking caveat in mind — it's the seam where VPNs and your Pi-hole rub against each other, which is exactly the Part 7 discussion.

The Tailscale exit node — a VPN built from your own tailnet

Before comparing products, here's the concept that makes Tailscale relevant to VPNs at all.

Normal Tailscale is an **overlay network**: it only carries traffic *between* your own devices (laptop `homelab` phone). Your ordinary internet traffic still leaves through your local connection; Tailscale doesn't touch it.

An **exit node** changes that. You designate one device on your tailnet as "the exit," and other devices route **all** their internet traffic through it — the full default route, not just tailnet traffic. To the outside world, your traffic now appears to come from the exit node's IP. That's exactly how a traditional full-tunnel VPN behaves, except the "VPN server" is a node on your own tailnet. Only one exit node is active per device at a time, and you switch it off by selecting "None."

This matters because Tailscale can use **Mullvad's servers as exit nodes** — giving you a real privacy VPN *inside* the same Tailscale app you already run, with no second VPN client. That's the key to making VPN and homelab coexist, and it's why Mullvad gets special attention below.

Your three real options

Option A — Tailscale's Mullvad add-on *(best fit for this homelab)*

Tailscale sells access to Mullvad's worldwide WireGuard servers as a paid add-on **purchased through Tailscale** — no separate Mullvad subscription. Mullvad's servers simply appear as selectable exit nodes inside your tailnet.

- **Cost:** about **$5/month**. Tailscale describes this as 5 "licenses," where one license covers up to 5 devices. (Tailscale's marketing page and its docs state the device math slightly differently — confirm the exact count at the checkout screen before buying.)
- **Enable it:** admin console → **Settings** → the **Mullvad** section → **Configure**, complete checkout, then grant your devices access.
- **Use it (Linux):**

  ```
  tailscale exit-node list                    # list Mullvad cities/servers
  tailscale set --exit-node=<mullvad-node> # route everything through it
  tailscale set --exit-node=                  # turn it back off
  ```

  On **iOS / macOS / Android**: the ••• menu → **Use exit node** → pick a Mullvad **location**.
- **Why it's the best fit here:** it *is* Tailscale, so it coexists with everything you've built — and crucially, it's the only option that works on your iPhone *without* fighting Tailscale (more on that in Part 7). The `--exit-node` choice is persistent across reboots, so "always on" is just setting it once.

Option B — A standalone VPN app (Aura, or the Mullvad app)

This is what you do today with **Aura**: a self-contained VPN app on each device that you toggle on to change location.

- **Aura:** part of the Aura identity-protection suite. A closed consumer app — easy to use, location-switching, with split tunneling on its supported platforms. It's fine for casual "make me look like I'm elsewhere." Its limitations for *this* project: it's app-only (there's no client you can run on the headless Pi, and no WireGuard config files to reuse), and like any full-device VPN it takes over DNS while it's on.
- **Mullvad app:** a flat **€5/month** (~$5.40), up to 5 devices, includes a proper **kill switch**, and — unlike Aura — Mullvad hands you **WireGuard config files** you can run anywhere, including on the Pi. That openness is what makes the advanced setups in Part 7 possible.

The catch with *any* standalone VPN app is how it coexists with Tailscale — a genuinely thorny topic that's the heart of Part 7. The short version: **phones run only one VPN at a time**, so the Aura/Mullvad app and Tailscale are mutually exclusive on your iPhone.

Option C — Your Pi as a self-hosted exit node (free, but not private)

You can advertise one of your *own* machines as an exit node:

```
sudo tailscale set --advertise-exit-node
# then approve it: admin console -> Machines -> the device ->
#   Edit route settings -> enable "Use as exit node"
```

On Linux you also enable IP forwarding (`net.ipv4.ip_forward=1` and the IPv6 equivalent).

- **What it's good for:** routing through your **home** Pi so you appear to be at home — reaching home-only services, or content tied to your home connection — and, importantly, it's the one exit option that keeps your traffic on your own network where **Pi-hole still applies**.
- **What it's not:** privacy. Traffic exits from **your own IP**, so there's no hiding. And don't use a *phone* as an exit node — it routes in userspace and is slow; the Pi (kernel routing) is fine.

Recommendation

For a homelab that's already built on Tailscale and a user who wants privacy *and* wants it to mesh with everything else:

- **For privacy / location changing: Option A (Tailscale's Mullvad add-on).** It's the same price ballpark as Aura or a standalone Mullvad sub, but it lives inside the one VPN your devices already run — so it's the only choice that works cleanly on your iPhone alongside the rest of the homelab, and the setting persists across reboots for true "always on."
- **Keep Option C (Pi as exit node) in your pocket** as a complement, not a replacement: flip to it when you want to appear at home or keep Pi-hole filtering active.

- **Aura (Option B) is fine to keep** as a standalone tool, but understand it won't integrate with the homelab — it can't run on the Pi, and on your phone it can't run at the same time as Tailscale. That trade-off, and how to live with it on the road, is precisely what Part 7 unpacks.

> ℹ "Always on" in one line
>
> With Option A, `tailscale set --exit-node=<mullvad-node>` (or picking a location in the mobile app) sticks across reboots and reconnects — you set it once per device. If the exit node ever becomes unreachable, traffic *fails closed* rather than leaking out your real connection, which is the behavior you want for an always-on privacy setup.

Recap

- A **VPN** hides your IP and shields your traffic from the local network — but it isn't anonymity and doesn't block ads on its own.
- A Tailscale **exit node** is a VPN built from your tailnet; with the **Mullvad add-on**, Mullvad's servers *are* your exit nodes, inside the app you already run.
- Three options: **A)** Tailscale's Mullvad add-on (best fit, works on iPhone), **B)** a standalone app like Aura or Mullvad's own, **C)** your Pi as a free-but-not-private exit node.
- **Recommendation:** use Option A for privacy, keep Option C for appear-at-home, and know Aura's limits before relying on it.

You now know your VPN options and how to turn each one on. The hard part — what actually happens to your Pi-hole, your homelab access, and your VPN when you walk out the front door, and how to get a setup that doesn't fight itself — is next, in Part 7.